USB-C, ESD

J1 USB-C

VBUS A4 · CC1 A5 · CC2 B5 · D− A7 · D− B7 · D+ A6 · D+ B6 · SBU1 A8 · SBU2 B8 · SHIELD · GND A1 · S1

VBUS, PWR_FLAG

C1 100n

USB_DM_C, USB_DP_C → U2 USBLC6-2SC6 → USB_DM, USB_DP

R1 5.1k, R2 5.1k

Shield straight to GND

Rd: USB-C sink

3V3 LDO

U3 AP7361C-33FGE

IN, EN, GND, OUT → +3V3

C2 2.2u, C3 10u

277 mA max, 0.62 W at 5.5 V in: EP to GND vias

RP2350A, FLASH, CRYSTAL

100 nF at IOVDD pins 1, 11, 20, 30, 38 (C4–C8), ADC_AVDD 44 and IOVDD 45 (C9, shared as in the vendor layout), USB_OTP_VDD 53 and QSPI_IOVDD 54 (C11). The Decouples field on each names its pin.

Regulator: C14 CIN and C15 COUT (0402 small pads), L1's polarity and copper all follow chips/rp2350.md

C15 COUT at L1; C16 at DVDD pin 23; C17, C18 at DVDD pins 6, 39

L1 3.3u

PWR_FLAG +1V1 → J3 1V1

C15 4.7u, C16 4.7u, C17 100n, C18 100n

C4 100n, C5 100n, C6 100n, C7 100n, C8 100n, C11 100n, C9 100n

R3 33R

C14 4.7u

C13 4.7u

VREG_AVDD, VREG_LX

U1 RP2350A

VREG_AVDD 46 · USB_OTP_VDD 53 · ADC_AVDD 44 · QSPI_IOVDD 54 · IOVDD 1 · VREG_VIN 49 · VREG_LX 48 · VREG_FB 50 · DVDD 6

RUN 26

27R at the chip

R4 27R USB_DM → USB_DM_MCU 51 · USB_DM

R5 27R USB_DP → USB_DP_MCU 52 · USB_DP

+3V3

C22 100n

U5 W25Q128JVSIQ

CS 1 · CLK 6 · DI/IO0 5 · DO/IO1 2 · WP/IO2 3 · HOLD/RESET/IO3 7 · GND 4 · VCC 8

QSPI_SS 60 · QSPI_SCLK 56 · QSPI_SD0 57 · QSPI_SD1 59 · QSPI_SD2 58 · QSPI_SD3 55

Y1 12MHz

C20 15p, C21 15p

XIN 21 · XOUT 22

R6 1k

SWCLK 24 · SWDIO 25

VREG_PGND, GND 61

GPIO0 2 · GPIO1 3 · GPIO2 4 · GPIO3 5 · GPIO4 7 · GPIO5 8 · GPIO6 9 · GPIO7 10 · GPIO8 12 · GPIO9 13 · GPIO10 14 · GPIO11 15 · GPIO12 16 · GPIO13 17 · GPIO14 18 · GPIO15 19 · GPIO16 27 · GPIO17 28 · GPIO18 29 · GPIO19 31 · GPIO20 32 · GPIO21 33 · GPIO22 34 · GPIO23 35 · GPIO24 36 · GPIO25 37 · GPIO26/ADC0 40 · GPIO27/ADC1 41 · GPIO28/ADC2 42 · GPIO29/ADC3 43

VREG_PGND joins GND at one point

GPIO HEADERS (TOP VIEW)

USB-C end

J2: GPIO0 1 · GPIO1 2 · 3 · GPIO2 4 · GPIO3 5 · GPIO4 6 · GPIO5 7 · 8 · GPIO6 9 · GPIO7 10 · GPIO8 11 · GPIO9 12 · 13 · GPIO10 14 · GPIO11 15 · GPIO12 16 · GPIO13 17 · 18 · GPIO14 19 · GPIO15 20 — GPIO left

J3: 20 VBUS · 19 GPIO29 · 18 · 17 GPIO28 · 16 +3V3 · 15 GPIO27 · 14 GPIO26 · 13 GPIO25 · 12 GPIO24 · 11 GPIO23 · 10 RUN · 9 GPIO22 · 8 · 7 GPIO21 · 6 GPIO20 · 5 GPIO19 · 4 GPIO18 · 3 · 2 GPIO17 · 1 GPIO16 — GPIO right

Pin map: parts/user-io.md. Not Pico pin-compatible: pins 31–39 differ. Headers are not assembled.

SWD, STATUS LED

J4 SWD: SWCLK 1 · GND 2 · SWDIO 3

R9 1k, D1 RED, GPIO25

RGB LEDS: VLED RAIL, LEVEL SHIFTER, 4 PIXELS

VBUS → D2 SM4007PL → VLED

R11 4.7k, D7 BZT52B5V1, C23 4.7u, C24 100n

PWR_FLAG

C25 100n, C26 100n, C27 100n, C28 100n

U4 SN74LV1T125DBVR (VCC, GND, OE, A, Y)

GPIO23 → R12 100k

R10 33R → LED_DIN

D3 WS2812B-2020-V6 (DIN, DOUT, VDD, VSS, Rgb) → PIX12 → D4 WS2812B-2020-V6 → PIX23 → D5 WS2812B-2020-V6 → PIX34 → D6 WS2812B-2020-V6

VLED 3.5–5.1 V: the rectifier's drop, held up by the 4.7k bleed, keeps the pixels inside 3.3–5.3 V; D7 clamps a hot-plug ring (parts/rgb-leds.md)

BUTTONS

R7 1k, QSPI_SS → SW1 BOOTSEL

R8 1k, RUN → SW2 RESET

Drawn by scripts/schematic.py	
Author: Sequoia Alexander	
CERN-OHL-P-2.0	
RP2350A dev board, 21 x 51 mm	
github.com/sequoia-hope/easy_pcb	
Sheet: /	
File: easy2350.kicad_sch	
Title: easy2350	
Size: A3	Date: 2026-10-23 — **Rev: A**
KiCad E.D.A. 9.0.8	Id: 1/1